

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Random Forest

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
100.0%	91.7%	0.7s

MODEL CONFIGURATION

Model Type	Random Forest
Number of Trees	100
Data Split	80% Training / 20% Validation
Extracted Features	mean, std, min, max, range, median
Feature Dimensions	6 statistics x 8 sensors = 48 features

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

POWER CONSUMPTION DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	6.92 W	Avg CPU Temp	47.3 °C
Peak Total Power	7.34 W	Peak CPU Temp	47.6 °C
Avg CPU+GPU Power	2.24 W	Energy Used	8.3 J (0.0023 Wh)
Peak CPU+GPU Power	2.63 W	Training Duration	1.0 s

MODEL EVALUATION RESULTS

Training Data — Accuracy: 100.00%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	1.0000	1.0000
Recall	1.0000	1.0000
F1-Score	1.0000	1.0000
Support (n samples)	96	96

Validation Data — Accuracy: 91.67%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9545	0.8846
Recall	0.8750	0.9583
F1-Score	0.9130	0.9200
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	21	3
Actual: Low	1	23

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

MODEL DETAILS & TRAINING CURVES

Feature Importance -- 100 Trees (Top 15)

No	Feature Name	Score (proportional bar)
1	tgs2600-range	 0.0607
2	tgs2620-range	 0.0482
3	tgs2600-std	 0.0411
4	tgs2600-mean	 0.0357
5	tgs2600-max	 0.0321
6	mq7-mean	 0.0321
7	mq7-max	 0.0321
8	mq3-mean	 0.0304
9	tgs2600-min	 0.0286
10	tgs2600-median	 0.0286
11	mq3-max	 0.0286
12	tgs2602-std	 0.0286
13	tgs2602-max	 0.0286
14	mq3-range	 0.0268
15	tgs2620-std	 0.0268

Accuracy Curve (Checkpoint)

N Trees	Training Accuracy (%)	Validation Accuracy (%)
10	97.92	93.75
20	98.96	93.75
30	99.48	93.75
40	99.48	91.67
50	99.48	91.67
60	100.00	91.67
70	100.00	91.67
80	100.00	91.67
90	100.00	91.67
100	100.00	91.67